

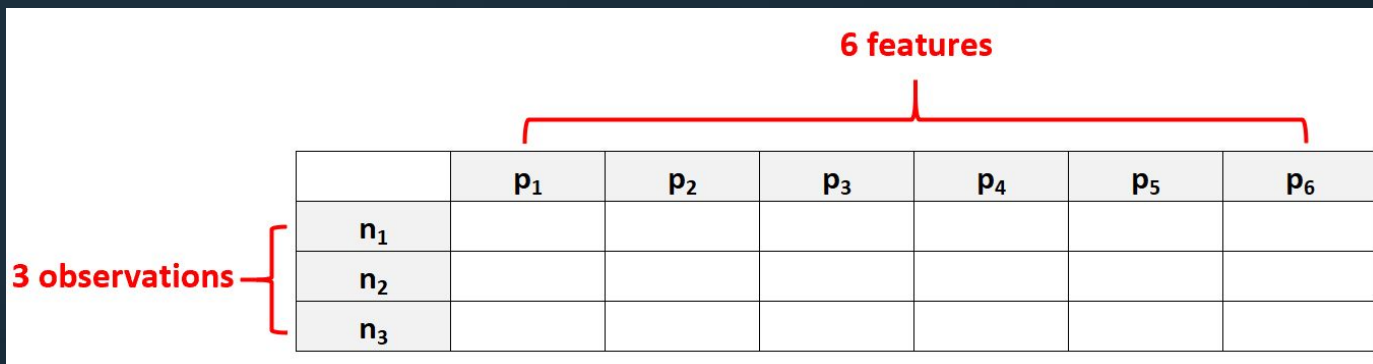
Dimension Reduction

Agenda

- What is Dimensionality?
- Need to be reduced ??
- Benefits of Dimension Reduction
- Feature Selection vs. Feature Extraction
- Intro to PCA

What is Dimensionality?

- *The number of input variables or features for a dataset*



A diagram illustrating a dataset with 3 observations and 6 features. The data is organized into a table with 7 columns and 4 rows. The first column contains observation labels n_1 , n_2 , and n_3 . The next six columns are labeled p_1 through p_6 , representing features. A red bracket on the left side of the table, spanning the three data rows, is labeled "3 observations". A red bracket on the top of the table, spanning the six feature columns, is labeled "6 features".

	p_1	p_2	p_3	p_4	p_5	p_6
n_1						
n_2						
n_3						

Need to be reduced ??

- Having a large number of dimensions in the feature space can mean that space is very large, and the points that we have in that space (rows of data) often represent a small and non-representative sample.
- This can impact the performance of machine learning algorithms fit on data with many input features, generally referred to as the “***curse of dimensionality***”
- Therefore, When dealing with high dimensional data, it is often useful to reduce the dimensionality by ***projecting the data to a lower dimensional subspace*** which ***captures the “essence” of the data***. This is called ***dimensionality reduction***.



Dimensionality Reduction

- Dimensionality reduction is a data preparation technique.
- It might be performed ***after data cleaning and data scaling and before training*** a predictive model.
- ***Your dataset will:***
 - be less complex
 - require less disk space
 - require less computation time
 - have lower chance of model overfitting
 - Can be easily visualized

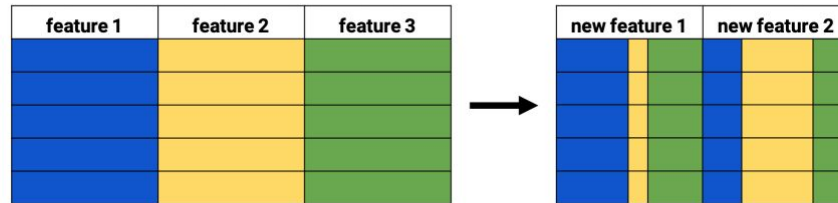
Dimensionality Reduction

- There are two types of Dimensionality Reduction techniques:

Feature selection



Feature extraction



Intro to PCA

- Transform to New Dimensions
- Capture Most of Variance (Information)

